

10.7.26

AI25BTECH11017-BALU

Question:

1) The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible values of the slope of this chord are: (2003)

- a) $-1, 1$
- b) $-2, 2$
- c) $-2, -\frac{1}{2}$
- d) $2, -\frac{1}{2}$

Solution:

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,

The parabola is

$$y^2 = 16x. \quad (1.1)$$

In quadratic form, this can be written as

$$\mathbf{x}^T \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -8 \\ 0 \end{pmatrix}^T \mathbf{x} = 0. \quad (1.2)$$

$$\mathbf{x}^T \mathbf{I} \mathbf{x} + 2 \begin{pmatrix} -6 \\ 0 \end{pmatrix}^T \mathbf{x} + 34 = 0. \quad (1.3)$$

Thus,

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = \begin{pmatrix} -6 \\ 0 \end{pmatrix}, \quad f = 34. \quad (1.4)$$

The focus of the parabola is

$$\mathbf{F} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}. \quad (1.5)$$

tangency condition

$$\mathbf{m}^T (\mathbf{q} \mathbf{v} + \mathbf{u}) = 0 \quad (1.6)$$

$$\mathbf{q} \text{ is point of contact} \quad (1.7)$$

$$(1.8)$$

equation of tangent

$$(\mathbf{q} \mathbf{v} + \mathbf{u})^T \mathbf{x} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (1.9)$$

$$\mathbf{m}^T(\mathbf{q}\mathbf{v} + \mathbf{u}) = 0 \quad (1.10)$$

$$(\mathbf{q}\mathbf{v} + \mathbf{u})^T \mathbf{F} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (1.11)$$

$$\mathbf{q}^T \mathbf{v} \mathbf{q} + 2 \begin{pmatrix} -6 \\ 0 \end{pmatrix}^T \mathbf{q} + 34 = 0. \quad (1.12)$$

by solving above equations we get

$$m = 1, -1$$

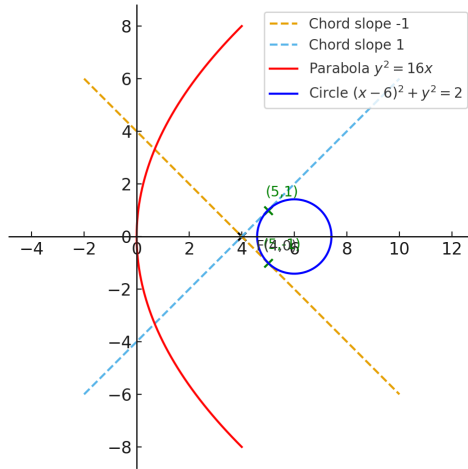


Fig. 1.1